

ANSYS

Fluent

Beginners to Advance level

✓ Quiz-2

- Enight → post processor

● Lesson 3-1

GE $\frac{\partial(\rho\phi)}{\partial t} + \text{div}(\rho\phi u) = \text{div}(\Gamma \text{grad}\phi) + S\phi$

for continuity $\phi = 1$

for momentum $\phi = \text{velocity}$

for Energy $\phi = h(\text{enthalpy})$

✓ Assignment 1

- ① Write down the governing equations of fluid dynamics used for CFD (continuity, NS and Energy) separately and explain the meaning / physics of each term.

⇒ Continuity Equation:

$$\underbrace{\frac{\partial \rho}{\partial t}}_{\text{transient term}} + \underbrace{\vec{\nabla} \cdot (\rho \vec{V})}_{\substack{\text{convective /} \\ \text{mass flux} \\ \text{term}}} = 0$$

fluid density velocity vector

for incompressible flows $\rho = \text{constant}$
then $\vec{\nabla} \cdot \vec{V} = 0$

volume flow entering a region equals the volume flow leaving it

Note Book

2

Sub.

Date

• NS equation

$$x \rightarrow \rho g_x - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) = \rho \frac{du}{dt}$$

$$y \rightarrow \rho g_y - \frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) = \rho \frac{dv}{dt}$$

$$z \rightarrow \rho g_z - \frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) = \rho \frac{dw}{dt}$$

Generally,

$$\rho \frac{D\vec{v}}{Dt} = -\nabla p + \rho \vec{g} + \mu \nabla^2 \vec{v}$$

$\downarrow$ total derivative
 $\underbrace{-\nabla p}_{\text{pressure gradient}}$
 $\underbrace{\rho \vec{g}}_{\text{body force term}}$
 $\underbrace{\mu \nabla^2 \vec{v}}_{\text{diffusion term}}$

$$\rho \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} \right]$$

$\downarrow$ change of velocity with time
 $\downarrow$ convective term

Rate of change
 of momentum
 = pressure forces
 + viscous forces
 + body forces

• Energy Equation

→ Energy is conserved

$$\rho \frac{d\hat{u}}{dt} + \rho (\vec{v} \cdot \vec{\nabla}) \hat{u} = \vec{\nabla} \cdot (k \vec{\nabla} T) + \phi$$

$\underbrace{\rho \frac{d\hat{u}}{dt}}_{\text{transient}}$
 $\underbrace{\rho (\vec{v} \cdot \vec{\nabla}) \hat{u}}_{\text{convective energy transport}}$
 $\underbrace{\vec{\nabla} \cdot (k \vec{\nabla} T)}_{\text{Heat conduction}}$
 $+ \phi$

(3)

Note Book

Sub.

Date

- ② Write down the general form of governing equation and explain the different terms in the equation

$$\underbrace{\frac{\partial}{\partial t}(\rho\phi)}_{\text{unsteady term}} + \underbrace{\vec{\nabla} \cdot (\rho \mathbf{u} \phi)}_{\text{convection term}} = \underbrace{\vec{\nabla} \cdot (\Gamma \vec{\nabla} \phi)}_{\text{diffusion term}} + \underbrace{S\phi}_{\text{source term}}$$

$\phi = 1 \rightarrow \text{continuity}$

$\phi = v \rightarrow \text{NS}$

$\phi = h \rightarrow \text{Energy}$

- ③ What are explicit and implicit schemes?
Give an example of an explicit scheme and implicit scheme.

➔ Explicit: Finding an unknown explicitly with the help of all knowns. Eg. FTCS

Implicit: No way to express the unknown with the help of knowns.
✓ eg. CRANK NICOLSEN

need to solve it using iterative method / matrix manipulation method

Explicit Scheme	Implicit scheme
1. Easy to setup and program	1. complicated to setup and program
2. Need to satisfy stability criteria hence Δt smaller for smaller Δx	2. Unconditionally stable

Note Book

④

Sub.

Explicit

Implicit

Date

3. computing time/time step less	3. larger due to matrix manipulation
----------------------------------	--------------------------------------

④ What is CFL number? Explain its significance

⇒

Courant - Friedrichs - Levy (non-dimensional number)

$CFL \leq 1$ for stable solution

$$\Rightarrow \frac{u \Delta t}{\Delta x} \leq 1$$

$u = vel$

$\Delta t = \text{timestep size}$

$\Delta x = \text{grid size}$

- For explicit time integration schemes, the CFL number must remain below the critical value

- It is widely used for convection diffusion problems

for implicit
 $CFL > 1$ is allowed and stable but costs accuracy

- primarily associated with hyperbolic equations

⑤ Write down the RANS equation and explain different terms in it

$$\Rightarrow \rho \left[\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right] = -\nabla \bar{p} + \nabla \tau_{ij}$$

transient term

$$\tau_{ij} = \mu \left(\frac{\partial \bar{u}_j}{\partial x_i} + \frac{\partial \bar{u}_i}{\partial x_j} \right) - \rho \bar{u}_i' \bar{u}_j'$$

laminar

turbulent Reynolds stress

pressure force produced due to mean pressure gradient

⑤

Note Book

Sub.

Date

- ⑥ Write down k and ϵ equation of the k - ϵ turbulence model and list down all the model constants.

$\Rightarrow$

Turbulence KE, $k = \frac{1}{2} \sum_{m=i,j,k} \overline{u'_m u'_m} = \frac{1}{2} (\overline{u'_1 u'_1} + \overline{u'_2 u'_2} + \overline{u'_3 u'_3})$

Turbulence eddy dissipation ϵ

$$= \nu \sum_m \sum_l \left(\frac{\partial u'_l}{\partial x_m} \right)^2$$

The transport equation for the turbulence KE and dissipation rate are:

1. $\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_j}(\rho \overline{v_j k}) = \underbrace{(\rho k - \rho \epsilon)}_{\text{turbulence production}} + \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right]$

2. $\frac{\partial}{\partial t}(\rho \epsilon) + \frac{\partial}{\partial x_j}(\rho \overline{v_j \epsilon}) = \epsilon_k (C_{\epsilon 1} \rho k - C_{\epsilon 2} \rho \epsilon) + \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right]$

C_μ	$C_{\epsilon 1}$	$C_{\epsilon 2}$	σ_k	σ_ϵ
0.09	1.44	1.92	1	1.3

$\rightarrow$ experimental

- ⑦ What is the law of wall and explain the concept of y^+

$\Rightarrow$ The law of wall is an empirical relationship that describes how the fluid velocity changes with distance from a solid wall in a turbulent boundary layer

y^+ is a dimensionless distance from wall

1. $y^+ < 5 \rightarrow$ viscous sublayer
2. $5 < y^+ < 30 \rightarrow$ buffer layer
3. $30 < y^+ < 300 \rightarrow$ log-law region

Sub.

Date

The value y^+ indicates where the first cell lies within the boundary layer

Higher $y^+ \rightarrow$ allows coarser mesh and lower computation cost by using wall functions

Lower $y^+ \rightarrow$ requires fine mesh, high computation cost

choosing appropriate y^+ is essential for obtaining accurate wall shear stress, drag and heat transfer predictions